

FRAME WORK OF ONLINE VOTING SYSTEM WITH FACE RECOGNATION USING BLOCK CHAIN

A PROJECT REPORT

Submitted by

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PANIMALAR ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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APPENDIX 2
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Certified that this project report “.....TITLE OF THE PROJECT...”

is the bonafide work of “.....NAME OF THE CANDIDATE(S).”

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Add ACKNOWLEDGEMENT TO OTHERS WHO HELPED IN YOUR PROJECT

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ABSTRACT

A widespread mistrust towards the traditional voting system has made democratic voting in any country very critical. People have seen their fundamental rights being violated. Other digital voting systems have been challenged due to a lack of transparency. Most voting systems are not transparent enough; this makes it very difficult for the government to gain voters' trust. The reason behind the failure of the traditional and current digital voting system is that it can be easily exploited. The primary objective is to resolve problems of the traditional and digital voting system, which include any kind of mishap or injustice during the process of voting. Blockchain technology can be used in the voting system to have a fair election and reduce injustice. The physical voting systems have many flaws in it as well as the digital voting systems are not perfect enough to be implemented on large scale. This appraises the need for a solution to secure the democratic rights of the people. This article presents a platform based on modern technology blockchain that provides maximum transparency and reliability of the system to build a trustful relationship between voters and election authorities. The proposed platform provides a framework that can be implemented to conduct voting activity digitally through blockchain without involving any physical polling stations. Our proposed framework supports a scalable blockchain, by using flexible consensus algorithms. The Chain Security Algorithm applied in the voting system makes the voting transaction more secure. Smart contracts provide a secure connection between the user and the network while executing a transaction in the chain. The security of the blockchain based voting system has also been discussed. Additionally, encryption of transactions using cryptographic hash and prevention of attack 51% on the blockchain has also been elaborated. Furthermore, the methodology for carrying out blockchain transactions during the process of voting has been elaborated using Blockchain. Finally, the performance evaluation of the proposed system shows that the system can be implemented in a large-scale population.

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LIST OF ABBREVIATIONS

JDK	Java Development Toolkit
DEX	Dalvik Executables
TCP	Transmission Control Protocol
IP	Internet Protocol
HTTP	Hyper Text Transfer Protocol
ADT	Android Development Tool

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CHAPTER 1

INTRODUCTION

Introduction:

Voting schemes have evolved from counting hands in early days to systems that include paper, punch card, mechanical lever and optical-scan machines. An electronic voting system which is used nowadays provide some characteristic different from the traditional voting technique, and also it provides improved features of voting system over traditional voting system such as accuracy, convenience, flexibility, privacy, verifiability and mobility. But Electronic voting systems suffers from various drawbacks such as time consuming, consumes large volume of paper work, no direct role for the higher officials, damage of machines due to lack of attention, mass update doesn't allows users to update and edit many item simultaneously etc. Therefore by implementing a decentralized Block chain based server environment we can prevent data loss.

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

This is the current voting system used in India. In this system vote is cast using electronics ballot. In this we cast our vote in an electronics machine. This is a group of some counter and registers. This voting system is quite easy, simple. It has advantage like mobility, secure, flexibility for election commission. But in today world all people are so much busy that they don't have time to vote. This paper presents a perspective in the electronic voting process. That includes but not limited to identifying the polling process, The polling process the actual voting process used on the polling day.

Disadvantage

The problems of the existing manual system of voting include among others the following:

- **Expensive and Time consuming:** The process of data collection and entry is costly and time-consuming. Expenses include printing forms, setting up registration stations, hiring personnel, and advertising registration days. Additionally, time is spent on voter sensitization and database entry
- **Too much paper work:** The process involves too much paper work and paper storage which is difficult as papers become bulky with the population size.
- **Short time provided to view the voter register:** This is a very big problem since not all people have free time during the given short period of time to check and update the voter register.

2.2 PROPOSED SYSTEM

The proposed Face Verified Online Voting System uses blockchain addresses for secure voter authentication. It verifies voter identity, allowing them to vote online. The voting process continues until the deadline, with real-time server updates. Votes are stored on a blockchain, ensuring transparency and trust.

Advantage

- Voter can cast their votes from anywhere in the country without visiting to voting booths, in highly secured way.
- This will increase the voting percentage in India and reduces the cost of voting process.
- By using Face Verification it provides enough security which reduces the false votes.

CHAPTER 3

REQUIREMENT SPECIFICATIONS

3.1 INTRODUCTION

Blockchain can reduce a lot of efforts and resources invested in polling stations, specifying the areas, appointing staff, and preventing security risks at polling stations [1]. Holding a digital election through blockchain not only saves money but also reduces the risk of inequity in the voting process [2]. Modern technologies such as blockchain technology are very secure and beneficial if used carefully. It can make the voting system more transparent, reliable, and also enhance traceability of transactions [3]. In a traditional digital voting system, a voting machine has been used which is connected to a centralized database. This machine can be tampered with by a person who has physical access to it. It may cause singlepoint of failure in the whole network of the voting system; on the other hand, an immutable blockchain would not be affected by an individual saboteur in the network [4]. In a blockchain, the data is being stored in a decentralized manner, which is constantly verified if the records are accurate. Therefore, in case of a malicious attack on a node, only that node would be affected and peer to peer network still provide all services. It makes blockchain technology a threatproofand reliable system to be used as a private ledger in the voting system. Blockchain has offered a level of security and trust more than the previously used technologies [5]. The resources can be saved by hiring less staff, security forces, and arranging polling stations for a traditional voting system can be given to the miners [6]. It does not require a physical module/voting machine which always has a chanceof cyber-attack through tampering devices [1]. The most ambient part of this system is the transparency which make people guarantee that their votes are being used in the right means. Blockchain technology is more secure and transparentthan any other database. It is being considered to be applied in critical systems like banking, medical _elds, food safety and cryptocurrency exchange [6]. Therefore, using Blockchain technology in the voting system assures that it will be more trustable, secure and consistent as compare to other digital voting systems. Blockchain technology can carry out a transparent voting process while maintaining voter privacy and trust.

3.2 HARDWARE AND SOFTWARE SPECIFICATION

3.2.1 HARDWARE REQUIREMENTS

- Hard Disk : 80GB and Above
- RAM : 4GB and Above
- Processor : P IV and Above

3.2.2 SOFTWARE REQUIREMENTS

- Windows 10 and above
- JDK 1.8
- Python 3.6.3
- XAMPP
- GanaChe

3.3 TECHNOLOGIES USED

- JAVA
- Blockchain

3.3.2 APACHE TOMCAT SERVER

Apache Tomcat, formerly part of the Apache Jakarta Project, is a web container developed by the Apache Software Foundation. It implements the servlet and JavaServer Pages (JSP) specifications, providing an environment for Java code to run with a web server. Tomcat includes tools for configuration and management, with XML-based configuration options. Since it has its own HTTP server, it can function as a standalone web server.

Environment

Tomcat is a web server that supports servlets and JSPs, featuring the Jasper compiler for JSP-to-servlet conversion. It can function independently or alongside other web servers like Apache. Once seen as suitable only for development, Tomcat is now widely used in high-traffic environments. Written in Java, it runs on any OS with a JVM.

Product features

Tomcat 3.x (initial release)

Implements the Servlet 2.2 and JSP 1.1 specifications

- implements the Servlet 2.2 and JSP 1.1 specifications
- servlet reloading
- basic HTTP functionality Tomcat 4.x
- implements the Servlet 2.3 and JSP 1.2 specifications
- servlet container redesigned as Catalina
- JSP engine redesigned as Jasper
- Coyote connector
- Java Management Extensions (JMX), JSP and Struts-based administration
- native Windows and Unix wrappers for platform integration
- faster JSP parsing

3.3.3 Blockchain Introduction:

With the rise of digital currency (cryptocurrency), enterprises are exploring Distributed Ledger technology to track asset ownership without a central authority. Blockchain, the core system behind this, consists of a chain of blocks secured using the SHA-256 hashing algorithm. Each block stores data and its hashed signature, ensuring security and integrity. Blockchain functions like a linked list, where each node (block) links to the previous one.

1. Block number – a sequence number (monotonically increasing) assigned to the block
2. Nonce – a random number which is used to generate Hash (as in #5) value which starts with 4 zeroes (0000). The process of generating this Nonce is called Mining.
3. Data – the actual user data associated with the block
4. Prev – contains the Hash of the previous block (e.g. current block # -1). The value for the first block in the chain is 64 zeroes (00).
5. Hash – current block's Hash value (generated using SHA-256). All of the above attributes excluding Hash e.g. Block #, Nonce, data, Prev are used to calculate the Hash of this block.

[#=1, Nonce=3409, Data=x, Prev=00..0, Hash=0000ffgr5rg67j] <- [#=2, Nonce=4986, Data=x, Prev=0000ffgr5rg67j, Hash=000045tggr5rg..77yh] <-.....and the chain goes on...

e.g. in above block #1, the value for Hash=0000ffgr5rg67j is generated using the values 1,3409,x,00..0. In case value for any of these 4 attributes changes, it will change the Hash value of this block. Once the Hash value of this Block changes (e.g. from 0000ffgr5rg67j to 34sdffgr5rg67j), it will break the next Block (#2) as its Prev field will point to invalid Hash (0000ffgr5rg67j doesn't exist anymore). This leads to a ripple effect and turns whole chain as invalid/tampered.

SHA256 Hash

Data:

test data

Hash:

916f0027a575074ce72a331777c3478d6513f786a591bd892da1a577bf2335f9

Block

Block: # 1

Nonce: 72608

Data:

Hash: 0000f727854b50bb95c054b39c1fe5c92e5ebcfa4bcb5dc279f56aa96a365e5a

Mine

Blockchain

Block: # 1

Nonce: 20839

Data: test data 1

Prev: 00

Hash: 00002a70c8d0034addeab115689ba9e79c8b8dbbd81b083be396c199bf

Mine

Block: # 2

Nonce: 81984

Data: test data 2

Prev: 00002a70c8d0034addeab115689ba9e79c8b8dbbd81b083be396c199bf

Hash: 0000a97e44b78fba8baabc2523d45e4a3cec092af26b185f29cd1336c94e

Mine

Block: # 3

Nonce: 79796

Data: test data 3

Prev: 0000a97e44b78fba8baabc2523d45e4a3c

Hash: 0000f2926342c369daff2bcc3fdad34f792

Mine

CHAPTER 4
Project Purpose and Scope

Purpose:

We are presenting a new e-Voting System with Face scanning that will overcome the drawbacks of the current voting methods that are used in India.

Project Scope:

The proposed system is a verified online voting system using Aadhaar-based face recognition. It identifies voters and allows only valid users to cast votes online. Voting continues until the set time ends, updating the database on the server. Aadhaar retrieves voter details, ensuring authenticity. Votes are stored on a blockchain server and made publicly viewable for transparency.

Product Perspective:

The current voting system in India is inefficient and vulnerable to threats, with voter ID cards often being faked. Manual vote counting is slow, and polling booths in rural areas face security issues, leading to ballot destruction. An online voting system can prevent such incidents and ensure more votes are counted. Blockchain technology enhances security and reliability by securely storing votes. This system eliminates fake voter IDs and speeds up the counting process.

System Features:

The electronic voting has emerged over time as a replacement to the paper-based voting to reduce the redundancies and inconsistencies. The historical perspective presented in the last two decades suggests that it has not been so successful due to the security and privacy observed over time. This paper suggests a framework by using effective hashing techniques to ensure the security of the data. The concept of block creation and block sealing is introduced in this paper.

System Design:

Voting schemes have evolved from counting hands in early days to systems that include paper, punch card, mechanical lever and optical-scan machines. An electronic voting system which is used nowadays provide some characteristic different from the traditional voting technique, and also it provides improved features of voting system over traditional voting system such as accuracy, convenience, flexibility, privacy, verifiability and mobility. Therefore by implementing a decentralized Blockchain based server environment we can prevent data loss.

4.5 Design and Implementation Constraints

4.5.1 Constraints in Analysis

- ◆ Constraints as Informal Text
- ◆ Constraints as Operational Restrictions
- ◆ Constraints Integrated in Existing Model Concepts
- ◆ Constraints as a Separate Concept

4.5.2 Constraints in Design

- ◆ Determination of the Involved Classes
- ◆ Determination of the Involved Objects
- ◆ Determination of the Involved Actions
- ◆ Determination of the Require Clauses

4.5.3 Constraints in Implementation

A hierarchical structuring of relations may result in more classes and a more complicated structure to implement. Therefore it is advisable to transform the hierarchical relation structure to a simpler structure such as a classical flat one. It is rather straightforward to transform the developed hierarchical model into a bipartite, flat model, consisting of classes on the one hand and flat relations on the other.

4.6 Other Nonfunctional Requirement

4.6.1 Performance Requirements

The application at this side controls and communicates with the following three main general components.

- embedded browser in charge of the navigation and accessing to the web service;
- Server Tier: The server side contains the main parts of the functionality of the proposed architecture. The components at this tier are the following.

Web Server, Security Module, Server-Side Capturing Engine, Preprocessing Engine, Database System, Verification Engine, Output Module.

4.6.2 Safety Requirements

1. The software may be safety-critical. If so, there are issues associated with its integrity level
2. The software may not be safety-critical although it forms part of a safety-critical system. For example, software may simply log transactions.
3. If a system must be of a high integrity level and if the software is shown to be of that integrity level, then the hardware must be at least of the same integrity level.
4. There is little point in producing 'perfect' code in some language if hardware and system software (in widest sense) are not reliable.
5. If a computer system is to run software of a high integrity level then that system should not at the same time accommodate software of a lower integrity level.
6. Systems with different requirements for safety levels must be separated.
7. Otherwise, the highest level of integrity required must be applied to all systems in the same environment.

CHAPTER 5

Architecture Diagram:

5.1 Architecture Diagram:

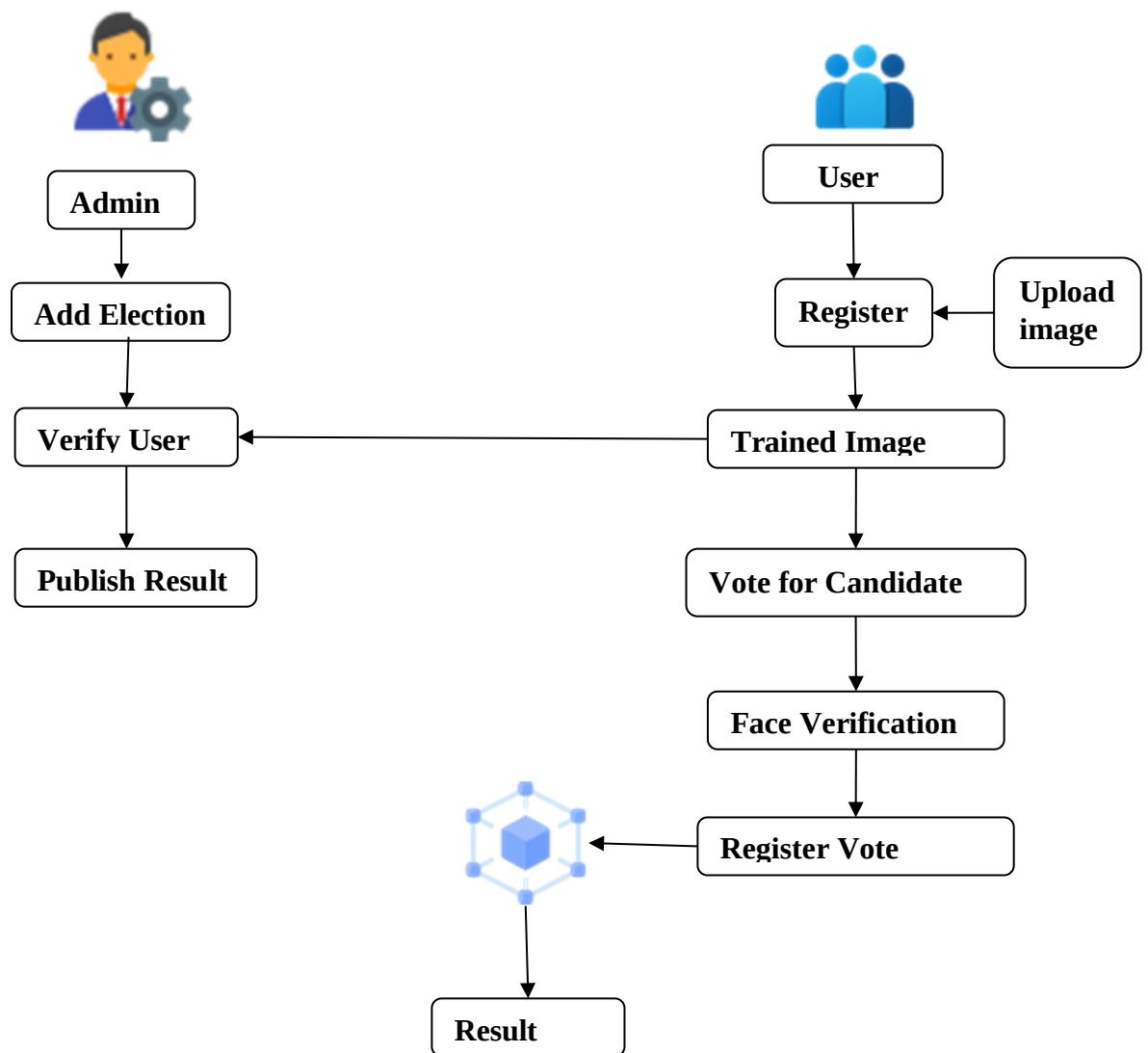
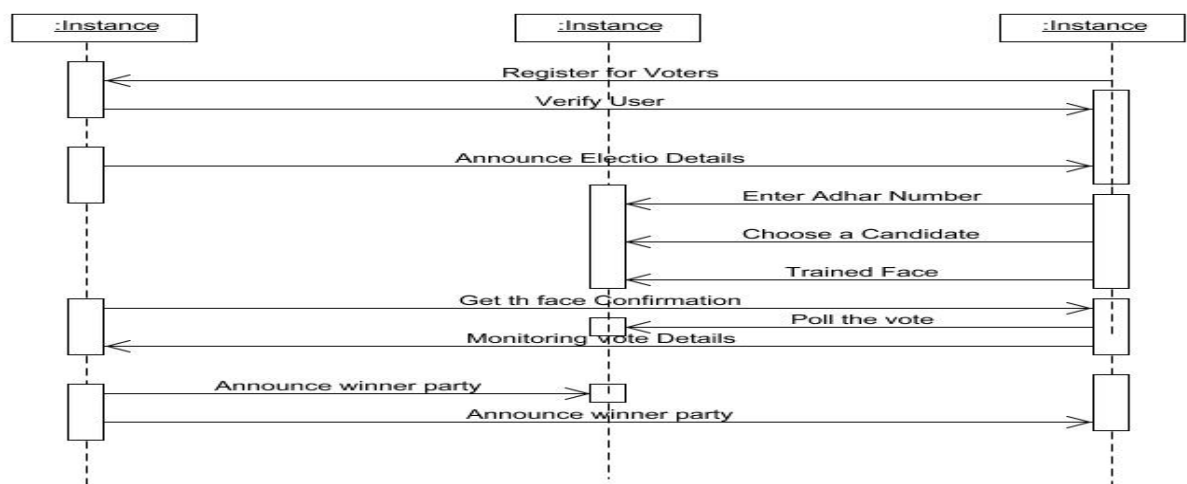


Fig: 5.1

5.2 Sequence Diagram:

A Sequence diagram is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of Message Sequence diagrams are sometimes called event diagrams, event sceneries and timing diagram.



5.3 Use Case Diagram:

Unified Modeling Language (UML) is a standardized general-purpose modeling language in the field of software engineering. The standard is managed and was created by the Object Management Group. UML includes a set of graphic notation techniques to create visual models of software intensive systems. This language is used to specify, visualize, modify, construct and document the artifacts of an object oriented software intensive system under development.

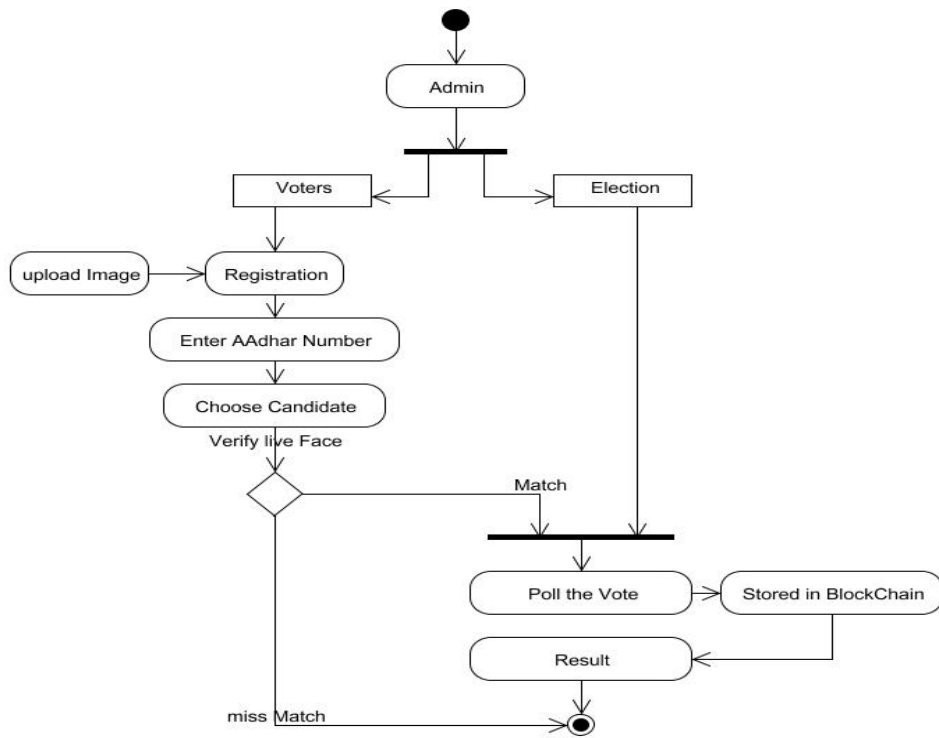
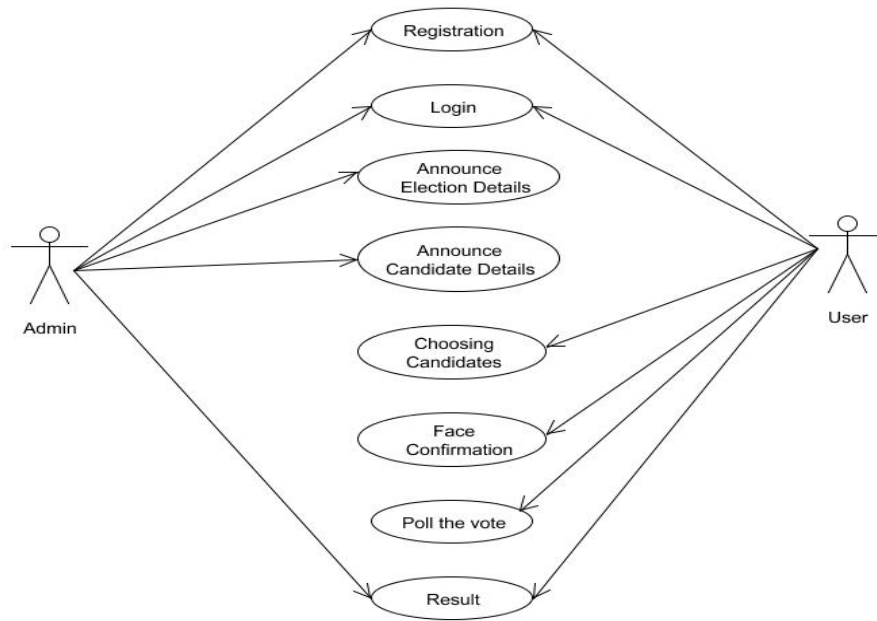
5.3.1. USECASE DIAGRAM

A Use case Diagram is used to present a graphical overview of the functionality provided by a system in terms of actors, their goals and any dependencies between those use cases.

Use case diagram consists of two parts:

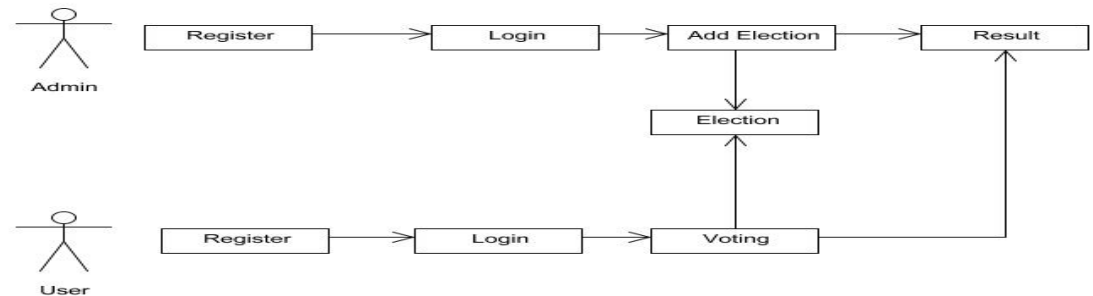
Use case: A use case describes a sequence of actions that provided something of measurable value to an actor and is drawn as a horizontal ellipse.

Actor: An actor is a person, organization or external system that plays a role in one or more interaction with the system.



5.5 Collaboration Diagram:

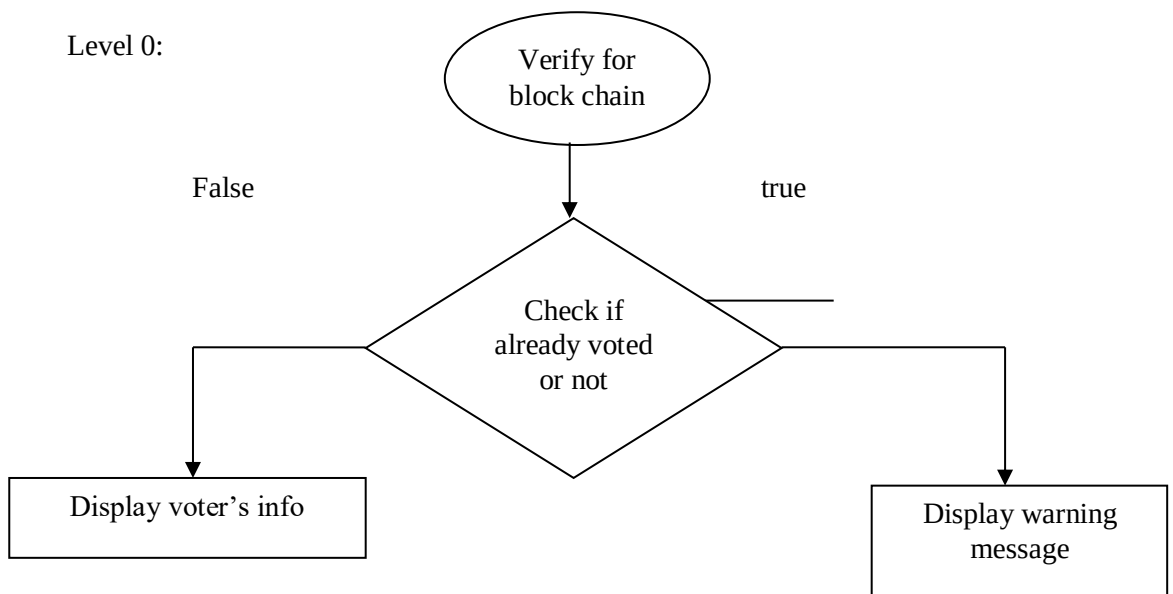
UML Collaboration Diagrams illustrate the relationship and interaction between software objects. They require use cases, system operation contracts and domain model to already exist. The collaboration diagram illustrates messages being sent between classes and objects.



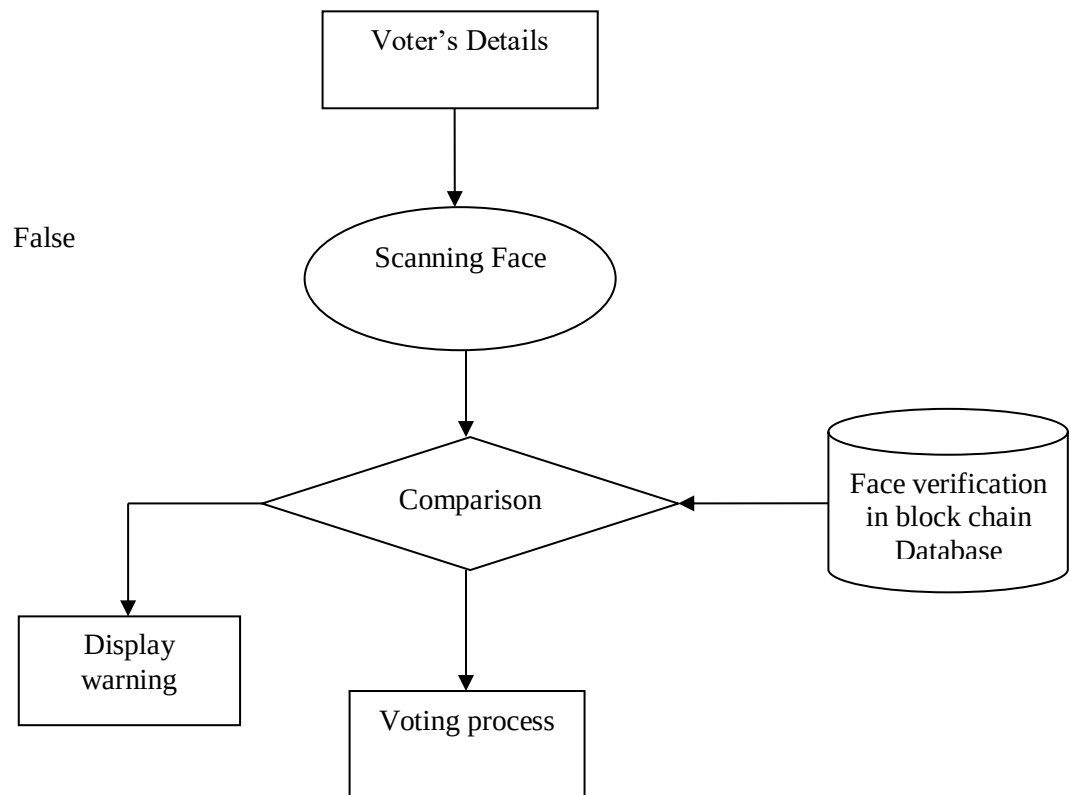
5.6 DATA FLOW DIAGRAM:

A Data Flow Diagram (DFD) is a graphical representation of the “flow” of data through an information system, modeling its aspects. It is a preliminary step used to create an overview of the system which can later be elaborated DFDs can also be used for visualization of data processing.

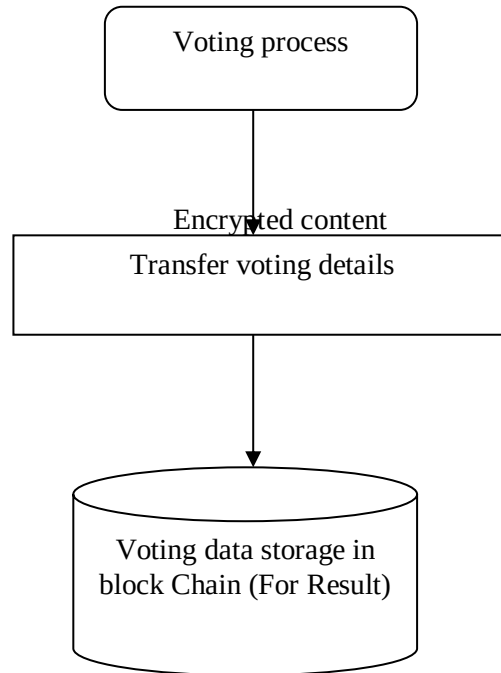
Level 0:



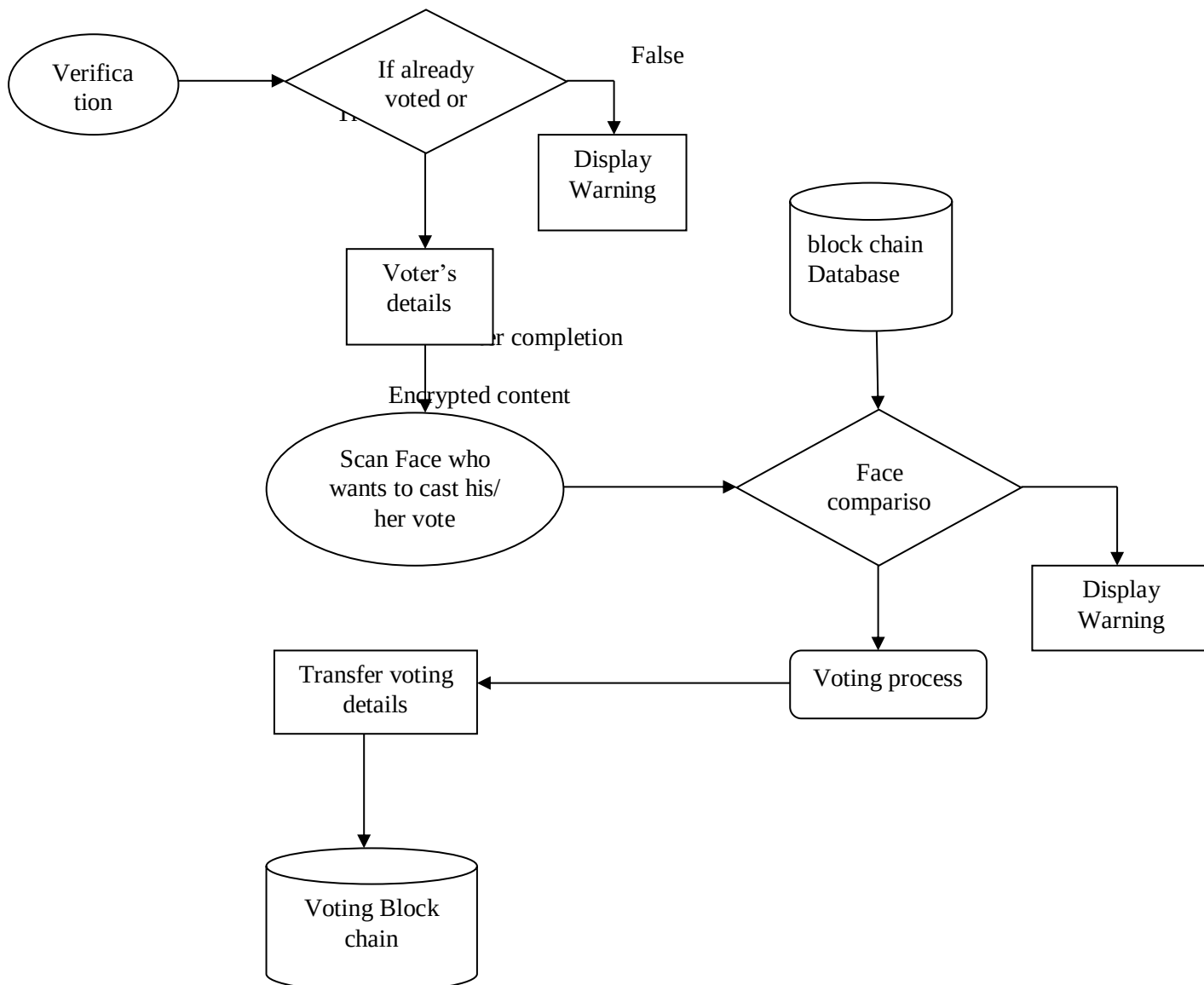
Level 1:



Level 2:



Level 3:



CHAPTER 6

MODULES

6.1 MODULES

- User Registration & Trained Voter Face
- Create Election
- Voting
- Publish Result

6.2 MODULE EXPLANATION:

6.2.1 1. User Registration & Trained Voter Face:

User should register in our website (*User Voting Page Way*) Block chain as an initial step with their mobile name, email, aadhar id, Voter id , image Area, Block chain Address contact number to which an unique *USER-ID* register. Users who are all registered in this portal are also considered as voter. The voter image convert to trained image After registering successfully the admin verify the voter details, after user can login into their profile using their *USER-ID* and their registered password.

1.Admin:

Admin Login page with default user name and password. Admin can accept or reject an voter request by verifying the user detail and also admin can register another admin. User has to scan his aadhar card for verification process. After scanning he should enter his detail and send an request to the admin if the account get rejected due to some reason he will be intimated to register again by admin.

2. Create Election

The Admin can create an election with election type and election constituency. All the election gets triggered at the given date and time. And Verified user has to login and scan his Block chain Address if election and user constituency matches user can view Election details. And Block chain Address. To create Nominated account in block chain.

3. Voting

Voters must have access to any web browser to take part in voting. The voter's interface would be provided in English language to make it easy to use for all users. The proposed system can contain a large number of voters at the time of voting. A decentralized block chain system enables a voter to vote from any part of the world. A

person can take part in voting from anywhere, even if he is in a foreign country, in this way his/her computerized National ID is verified from the national database so he can cast the vote. User has to face his registered finger during his registration process. In voting page voter has to scan his face if the User Face matches with registered Face, voter can cast his or her vote to the right candidate. Source KNN an algorithm for recognition of human face is used to compare two Face. Voting transactions are sent to a pool from which miners analyze them and remove the malicious request by taking the consensus from the other nodes before adding it to the chain. The votes are fully secured using a cryptographic hash. Each vote cast adds a new block in the chain. When the transaction completes and a node is successfully added to Vote Chain, the voter of that particular voting transaction is notified through an SMS to his registered email. The voter has provided with a unique transaction hash by which he can verify his vote through a web portal and upon successful completion of transaction the vote has been counted in the whole voting activity.

4. Publish Result

Smart contracts are providing a secure connection between the user and the network while executing a transaction in the chain. These are the rules that are implemented on the entire Block chain and cannot be neglected under any condition. All the nodes have to follow the smart contracts to save the vote in the system successfully. When user completes his or her voting process votes are stored in Block chain. So the voter can trust his votes stored in block chain cannot be changed. User can view his or her vote in a pie chart retrieved from block chain. SHA256 algorithm has been used to hash the data. Admin Can publish the result of each constituency after the election process is fully completed.

CHAPTER 7

CODING AND TESTING

7.1 CODING

Once the design aspect of the system is finalized the system enters into the coding and testing phase. The coding phase brings the actual system into action by converting the design of the system into the code in a given programming language. Therefore, a good coding style has to be taken whenever changes are required it easily screwed into the system.

7.2 CODING STANDARDS

Coding standards are guidelines to programming that focuses on the physical structure and appearance of the program. They make the code easier to read, understand and maintain.

Program should be simple, clear and easy to understand.

Naming conventions

Value conventions

Message box format

Exception and error handling

7.2.1 NAMING CONVENTIONS

Naming conventions of classes, data member, member functions, procedures etc., should be **self-descriptive**. One should even get the meaning and scope of the variable by its name. The conventions are adopted for **easy understanding** of the intended message by the user. So it is customary to follow the conventions. These conventions are as follows:

Class names

Class names are problem domain equivalence and begin with capital letter and have mixed cases.

Member Function and Data Member name

Member function and data member name begins with a lowercase letter with each subsequent letters of the new words in uppercase and the rest of letters in lowercase.

7.2.3 SCRIPT WRITING AND COMMENTING STANDARD

Script writing is an art in which indentation is utmost important. Conditional and looping statements are to be properly aligned to facilitate easy understanding. Comments are included to minimize the number of surprises that could occur when going through the code.

7.2.4 MESSAGE BOX FORMAT

When something has to be prompted to the user, he must be able to understand it properly. To achieve this, a specific format has been adopted in displaying messages to the user. They are as follows:

- X – User has performed illegal operation.
- ! – Information to the user.

7.3 TEST PROCEDURE SYSTEM TESTING

Testing is performed to identify errors. It is used for quality assurance. Testing is an integral part of the entire development and maintenance process. The goal of the testing during phase is to verify that the specification has been accurately and completely incorporated into the design, as well as to ensure the correctness of the design itself. For

- Static analysis is used to investigate the structural properties of the Source code.
- Dynamic testing is used to investigate the behavior of the source code by executing the program on the test data.

7.4 TEST DATA AND OUTPUT

7.4.1 UNIT TESTING

Unit testing is conducted to verify the functional performance of each modular component of the software. Unit testing focuses on the smallest unit of the software design (i.e.), the module. The white-box testing techniques were heavily employed for unit testing.

7.4.2 FUNCTIONAL TESTS

Functional test cases involved exercising the code with nominal input values for which the expected results are known, as well as boundary values and special values, such as logically related inputs, files of identical elements, and empty files.

Three types of tests in Functional test:

- Performance Test
- Stress Test
- Structure Test

7.4.3 PERFORMANCE TEST

It determines the amount of execution time spent in various parts of the unit, program throughput, and response time and device utilization by the program unit.

7.4.5 STRUCTURED TEST

Structure Tests are concerned with exercising the internal logic of a program and traversing particular execution paths. The way in which White-Box test strategy was employed to ensure that the test cases could Guarantee that all independent paths within a module have been exercised at least once.

- Exercise all logical decisions on their true or false sides.
- Execute all loops at their boundaries and within their operational bounds.
- Exercise internal data structures to assure their validity.

7.4.6 INTEGRATION TESTING

Integration testing is a systematic technique for construction the program structure while at the same time conducting tests to uncover errors associated with interfacing. i.e., integration testing is the complete testing of the set of modules which makes up the product. The objective is to take untested modules and build a program structure tester should identify critical modules. Critical modules should be tested as early as possible. One approach is to wait until all the units have passed testing, and then combine them and then tested.

7.5 TESTING TECHNIQUES / TESTING STRATERGIES

7.5.1 TESTING

Testing is a process of executing a program with the intent of finding an error. A good test case is one that has a high probability of finding an as-yet –undiscovered error. A successful test is one that uncovers an as-yet- undiscovered error. System testing is the stage of implementation, which is aimed at ensuring that the system works accurately and efficiently as expected before live operation commences. It verifies that the whole set of programs hang together. System testing requires a test consists of several key activities and steps for run program, string, system and is important in adopting a successful new system.

The software testing process commences once the program is created and the documentation and related data structures are designed. . Testing is the process of executing the program with the intent of finding the error. A good test case design is one that as a probability of finding an yet undiscovered error. A successful test is one that uncovers an yet undiscovered error. Any engineering product can be tested in one of the two ways:

7.5.1.1 WHITE BOX TESTING

This testing is also called as Glass box testing. In this testing, by knowing the specific functions that a product has been design to perform test can be conducted that demonstrate each function is fully operational at the same time searching for errors in each function. It is a test case design method that uses the control structure of the procedural design to derive test cases. Basis path testing is a white box testing.

Basis path testing:

- Flow graph notation
- Cyclometric complexity
- Deriving test cases
- Graph matrices Control

7.5.1.2 BLACK BOX TESTING

In this testing by knowing the internal operation of a product, test can be conducted to ensure that “all gears mesh”, that is the internal operation performs according to specification and all internal components have been adequately exercised. It fundamentally focuses on the functional requirements of the software.

The steps involved in black box test case design are:

- Graph based testing methods
- Equivalence partitioning
- Boundary value analysis
- Comparison testing

7.5.2 SOFTWARE TESTING STRATEGIES:

A software testing strategy provides a road map for the software developer. Testing is a set activity that can be planned in advance and conducted systematically. For this reason a template for software testing a set of steps into which we can place specific test case design methods should be strategy should have the following characteristics:

- Testing begins at the module level and works “outward” toward the integration of the entire computer based system.
- The developer of the software and an independent test group conducts testing.
- Testing and Debugging are different activities but debugging must be accommodated in any testing strategy.

7.5.2.1 INTEGRATION TESTING:

Integration testing is a systematic technique for constructing the program structure while at the same time conducting tests to uncover errors associated with. Individual modules, which are highly prone to interface errors, should not be assumed to work instantly when we put them together.

7.5.2.2 PROGRAM TESTING:

The logical and syntax errors have been pointed out by program testing. A syntax error is an error in a program statement that in violates one or more rules of the language in which it is written. An improperly defined field dimension or omitted keywords are common syntax error.

7.5.2.3 SECURITY TESTING:

Security testing attempts to verify the protection mechanisms built in to a system well, in fact, protect it from improper penetration. The system security must be tested for invulnerability from frontal attack must also be tested for invulnerability from rear attack.

7.5.2.4 VALIDATION TESTING

At the culmination of integration testing, software is completely assembled as a package. Interfacing errors have been uncovered and corrected and a final series of software test-validation testing begins. Validation testing can be defined in many ways, but a simple definition is that validation succeeds when the software functions in manner that is reasonably expected by the customer. Software validation is achieved through a series of black box tests that demonstrate conformity with requirement.

* The function or performance characteristics confirm to specifications and are accepted.

* A validation from specification is uncovered and a deficiency created.

7.5.2.5 USER ACCEPTANCE TESTING

User acceptance of the system is key factor for the success of any system. The system under consideration is tested for user acceptance by constantly keeping in touch with prospective system and user at the time of developing and making changes whenever required. This is done in regarding to the following points.

- Input screen design.
- Output screen design.

Source Code


```
//Voters
<?PHP
require "util.php";
//session_start();
/*
$image;
$aadhar;
$name;
$mobile;
$dob;
$check;
$number;
$status="Pending";
$password;
$error = Array();
if(isset($_POST['submit'])){
    //$image=$_POST['vimage'];
    $img = $_POST['image'];
    $aadhar=$_POST['vaadhar'];
    $name=$_POST['vname'];
    $id=$_POST['vid'];
    $dob=$_POST['dob'];
    //$fpstr=$_POST['b64img'];
    $area=$_POST['area'];
    $password = $_POST['password'];
    //$email = $_POST['email'];
    $check = strlen($aadhar);
    $ids = strlen($id);
    $baddress = $_POST['baddress'];

    // echo "SASDASD ";
    // echo "image".$img;
    //echo $name===="";
    if(!isset($img) || $img==''){
```

```

// echo "Says";
$error['image']='Please Upload an image';
}
if(!isset($aadhar) || $aadhar==''){
    $error['aadhar']='Please Enter the Aadhar Id';
}

    if(!isset($password) || $password==''){
        $error['password']='Please Enter the pssword';
    }

    $email = $_POST["email"];
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $emailErr = "Invalid email format";
    }

    if($check<12 || $check>12 || !is_numeric($aadhar)){
        $error['aadhar']='Please Enter valid 12 Digit Aadhar Id';
    }

    if($ids>10 || $ids<10 || !is_numeric($id)){
        $error['id']='Plese Enter valid 10 Digit Voter Id';
    }

if(!isset($name) || $name==''){
    $error['name']='Please Enter the name';
}

if(!isset($id) || $id==''){
    $error['id']='Please Enter the Voter ID';
}

if($dob<18)
{
    $error['dob']='18 plus aged people are only eligible for elections';
}

    if(!isset($baddress) || $baddress==''){
        $error['id']='Please Enter the Blockchain Address';
    }

}

/*****/
/* $folderPath = "voterImages/";

```

```

$image_parts = explode(";base64,", $img);
$image_type_aux = explode("image/", $image_parts[0]);
$image_type = $image_type_aux[1];
$image_base64 = base64_decode($image_parts[1]);
$fileName = uniqid() . '.png';
$file = $folderPath . $fileName;
file_put_contents($file, $image_base64);
if(sizeof($error)==0)
{
$fingerimage="d16e7d9f1e1c68c49b30c9364d3e0b23.png";
$sql="insertintovoterinfo(vimage,vaadhar,vname,password,email,vid,dob,area,status,finger,baddress)"
"values('".$fileName."','".$aadhar."','".$name."','".$password."','".$email."','".$
$.id."','".$dob."','".$area."','".$status."','".$fingerimage."','".$baddress."')";
header('Location: ../voters.php');
if($db->query($sql)){
$success="Voter Added Successfully ...Waiting for Approval";
}
else{
$er = $db->error;
//echo $er;
unlink($fpath);
unlink( $target_path);
if(strpos($er, 'key 3') !== false){
$error['ER_EM']='Voter id already registered';
}
elseif ((strpos($er, 'key 2') !== false)) {
$error['ER_EM']='Aadhar id already registered';
}
else{
$error['ER_EM']=$er;;
}
}
}
}

```

```

    }
    else{
        $error=[];
    }
*/
?>
<!DOCTYPE HTML>
<html>
<head>
<title>Election a Society and People Category </title>
<!-- for-mobile-apps -->
<meta name="viewport" content="width=device-width, initial-scale=1">
<meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/2.1.1/jquery.min.js"></script>
<script type="application/x-javascript"> addEventListener("load", function() {
setTimeout(hideURLbar, 0); }, false);
function hideURLbar(){ window.scrollTo(0,1); } </script>
<!-- //for-mobile-apps -->
<link href="assets/css/bootstrap.css" rel='stylesheet' type='text/css' />
<!--<link
href='//fonts.googleapis.com/css?family=Raleway:400,100,200,300,500,600,700,80
0,900' rel='stylesheet' type='text/css'>-->
<!--<link
href='//fonts.googleapis.com/css?family=Open+Sans:400,300,300italic,400italic,6
00,600italic,700,700italic,800,800italic' rel='stylesheet' type='text/css'>-->
<link rel="stylesheet" type="text/css" href="assets/css/style.css">
<!--strat-slider---->
<!--script type="text/javascript" src="assets/js/jquery-1.11.1.min.js"></script-->
<script src="https://code.jquery.com/jquery-2.2.4.min.js" integrity="sha256-
BbhdLvQf/xTY9gja0Dq3HiwQF8LaCRTXxZKRutelT44="
crossorigin="anonymous"></script>
<!--//slider---->
<!--scripttype="text/javascript" src="assets/js/adapter.min.js"></script>

```

```

<script type="text/javascript" src="assets/js/vue.min.js"></script>
<script type="text/javascript" src="assets/js/instascan.min.js"></script-->
<script src="https://cdnjs.cloudflare.com/ajax/libs/webcamjs/1.0.25/webcam.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/web3/1.8.1/web3.in.js"></script>
<script src="https://cdn.jsdelivr.net/npm/truffle-contract@4.0.16/dist/truffle-contract.js"></script>
<script src="mfs100-9.0.2.6.js"></script>
<script type="text/javascript" src="config.js"></script>
<script language="javascript" type="text/javascript">
var datauri="";

    function processResult(str){
        //alert(str);
        $xmlDoc = $.parseXML( str );
        uid = $xmlDoc.children[0].getAttribute('uid');
        $("#aid").val(uid);
        uid = $xmlDoc.children[0].getAttribute('name');
        $("#vname").val(uid);
        uid = $xmlDoc.children[0].getAttribute('dob');
        var parts =uid.split('/');
        var mydate = new Date(parts[2], parts[1] - 1, parts[0]);
        $("#dob").val(mydate);
    }
</script>
<script>
    var _MS_PER_DAY = 1000 * 60 * 60 * 24;
function chkForm(){
    var a  = new Date();
    dateStr = document.getElementById("dob").value;
    b =new Date(dateStr);
    //alert(b);
    //var b  = new Date("2017-07-25");
    var remainingDays  = dateDiffInDays(b, a);

```

```

if(remainingDays<6570){
    alert("Age should be above 18 yrs");
    return false;
}
return true;
}

function dateDiffInDays(a, b) {
    // Discard the time and time-zone information.
    var utc1 = Date.UTC(a.getFullYear(), a.getMonth(), a.getDate());
    var utc2 = Date.UTC(b.getFullYear(), b.getMonth(), b.getDate());
    return Math.floor((utc2 - utc1) / _MS_PER_DAY);
}
</script>
</head>
<body>
<!-- header -->
<div class="header_bg">
<div class="container">
<!--start-header-->
<div class="header">
<div class="logo">
<!--    <ahref="#"></a> -->
</div>
<nav class="navbar navbar-default">
<!-- Brand and toggle get grouped for better mobile display -->
<div class="navbar-header">
<button type="button" class="navbar-toggle collapsed" data-toggle="collapse"
data-target="#bs-example-navbar-collapse-1">
<span class="sr-only">Toggle navigation</span>
    <span class="icon-bar"></span>
    <span class="icon-bar"></span>
    <span class="icon-bar"></span>
  </button>
</div>

```

```

<!-- Collect the nav links, forms, and other content for toggling -->
<div class="collapse navbar-collapse nav-wil" id="bs-example-navbar-collapse-
1">
<ul class="nav navbar-nav">
  <li><a href="index.php">Home</a></li>
  <!--<li><a href="admin.php">Login</a></li>-->
  <li><a href="voting.php">Voting</a></li>
<li><a href="user.php">User Login</a></li>
<li class="act"><a href="voters.php">User Register</a></li>
  <li><a href="results.php">Results</a></li>
</ul>
</div><!-- /.navbar-collapse -->
</nav>
</div>
</div>
</div>
<div class="header_bottom">
</div>
<!-- //end-header -->
<!-- banner1 -->
<!-- //banner1 -->
<!--typography-page -->
<div class="typo">
<form action="#" method="post" id="myform" onsubmit="return
chkForm();" enctype="multipart/form-data">
<div class="container">
<h3 class="title">Welcome for Online Voting</h3>
<p class="nihil">Vote For Real Government.</p>
<div class="grid_3 grid_4">
<h3 class="hdg">Voter Citizen Details</h3>
<div id="response" class="alert alert-danger collapse" role="alert">
  </div>
<div id="success" class="alert alert-success collapse" role="alert">
  </div>

```

```

<div class="bs-example" style="width:50%;float:left">
<table class="table">
<tbody>
  <tr>
    <td>
      <!--?php
      foreach ($error as $value) {
        echo "<span style='color:red'>$value</span><br/>";
      }
      ?>
    <!--?php
    foreach ($success as $value) {
      echo "<span style='color:green'>$value</span><br/>";
    }
    ?-->
  </td>
</tr>
<tr>
  <td><h4id="h4.-bootstrap-heading"><b>Voter    Image</b></h4><br/><input
type="file" name="vimage" onchange="previewFile()" style="border: none;
border-bottom: 2px solid steelblue;width: 80%;height:50px" /></td>
</tr>
<tr>
  <td>
<h4id="h4.bootstrapheading"><b>VoterAdharNumber</b></h4><input
type="text" name="vaadhar" id="aid" style="border: none; border-bottom:
2px solid steelblue;width: 80%;height:50px" value="<?php echo isset($aadhar)
? $aadhar:"" ?>"/></td>
</tr>
<tr>
  <td><h4id="h4.bootstrapheading"><b>VoterName</b></h4><input
type="text" name="vname" id="vname" style="border: none; border-bottom:
2px solid steelblue;width: 80%;height:50px" value="<?php echo isset($name)?
$name : "" ?>"/></td>

```



```

</tr>
<tr>
<td>
<h4id="h4.-bootstrapheading"><b>VoterPassword</b></h4><input
type="password" name="password" id="passwrđ" style="border: none;
border-bottom: 2px solid steelblue;width: 80%;height:50px" value="<?php
echo isset($password)? $password : "" ?>" /></td>
</tr>
<tr>
<td><h4id="h4.-bootstrap-heading"><b>Voter Email</b></h4><input
type="text" name="email" id="email" style="border: none; border-bottom:
2px solid steelblue;width: 80%;height:50px" value="<?php echo isset($email)?
$email : "" ?>" /></td>
</tr>
<tr>
<td><h4 id="h4.-bootstrap-heading"><b>Voter Id</b></h4><input
type="text" name="vid" style="border: none; border-bottom: 2px solid
steelblue;width: 80%;height:50px" value="<?php echo isset($id)?$id:""
?>" /></td>
</tr>
<tr>
<td><h4 id="h4.-bootstrap-heading"><b>Area</b></h4>
<select name="area" id="subtype" style="border: none; border-bottom: 2px
solid steelblue;width: 80%;height:50px" required>
<option>-----CHOOSE HERE-----</option>
<?php
$sql="select distinct area from constituency";
$ret = $db->query($sql);
//echo $sql;
while($row = $ret->fetch_assoc()) {
echo"<option value=\"\".\".$row['area'].\"\">\".$row['area'].\"</option>\";
} ?>
<!--option value="Alandur">By Election</option-->
</select>

```

```

        </td>
    </tr>
    <tr>
        <td><h4 id="h4.-bootstrap-heading"><b>Voter Date Of Birth</b></h4><input
        type="date" name="dob" id="dob" data-date-inline-picker="true"
        style="border: none; border-bottom: 2px solid steelblue;width:
        80%;height:50px" value="<?php echo isset($dob)?$dob:"" ?>" /></td>
    </tr>
    <tr>
        <td><h4 id="h4.-bootstrap-heading"><b>Blockchain Address</b></h4><input
        type="text" name="baddress" style="border: none; border-bottom: 2px solid
        steelblue;width: 80%;height:50px" value="<?php echo isset($baddress)?
        $baddress : "" ?>" /></td>
    </tr>
</tbody>
</table>
</div>
<div lass="grid_3 grid_5" style="width:40%;float:right" tyle="margin-left:
770px;float: left;margin-top: -260px">
    <div id="app" style="width:100px">
        <section class="cameras">
            <h3>Cameras</h3>
            <div id="my_camera"></div>
            <div id="container">
                <!--video width="200" height="200" autoplay="true" id="video">
            </video-->
                <!--inputtype=buttonvalue="TakeSnapshot"onClick="take_snapshot()" -->
                <input type="button" id="btnCapture" value="Capture" class="
                capturebuttonpaddingbtnbtnprimarybtnlgsubmit_bottom_padding"
                onClick="take_snapshot()" style="margin-top:20px"/>
                <input type="hidden" name="image" class="image-tag">
            </div>
            <br> </br>
            <br/><br/>
</section>
</div>
</div>
</div>
<p class="one"><input type="button" id="submit" name="submit" value="
ADD VOTER DETAILS" style="background-color:steelblue;color:white
;height: 50px;width: 100%;border: 2px solid steelblue;"></p>
</div>
</form>
<!-- /.row -->
</div>
<!-- footer -->
<!-- //footer -->
<!-- scroll_top_btn -->
<script type="text/javascript" src="assets/js/move-top.js"></script>
<script type="text/javascript" src="assets/js/easing.js"></script>
<script type="text/javascript">
$(document).ready(function() {
//$("#success").toggleClass("hidden");
//$("#response").toggleClass("hidden");
var defaults = {
containerID: 'toTop', // fading element id
containerHoverID: 'toTopHover', // fading element hover id
scrollSpeed: 1200,
easingType: 'linear'
};
$.UItoTop({ easingType: 'easeOutQuart' });
});
</script>
<script>
$('#submit').click(function(event){
//$("#success").toggleClass("hidden");
//$("#response").toggleClass("hidden");

```

```

event.preventDefault();
var myFormData = new FormData($(this).parents('form')[0]);
$("#response").empty();
jQuery('#response').html('');
$.ajax({
url: 'addvoter.php',
type: 'POST',
processData: false, // important
contentType: false, // important
datatype : "application/json",
//contentType: "text/plain"
data: myFormData,success: function(result)
{
res = $.parseJSON(result);
console.log(res);
console.log(res['stat']);
//$("#response").empty();
if (res['stat']==true)
{
Addvoterface();
console.log("inside true");
// $("#success").toggleClass("hidden");
// $("#response").toggleClass("hidden");
$('#response').hide();
$('#success').show();
//var element = document.getElementById("response");
//element.style.color = "#008000";
for (var i in res) {
if (i !='stat')
{
$("#success").append(res[i]+ " "+"<br/>");
}
}
}
$('#myform')[0].reset();

```

```

//$(this).closest('myform').find("input[type=text], textarea").val(""); }
else{
    /*if ($("#success").is(":hidden")) {}
else{
    $("#success").toggleClass("hidden");
    }*/
    //if ($("#response").is(":hidden")) {
    // $("#response").toggleClass("hidden");
    //}
    $('#response').show();
    $('#success').hide();
    //var element = document.getElementById("response");
    //element.style.color = "#FF0000";
    for (var i in res) {
        if (i != 'stat')
        {
            //$("#response").append(i+ ":" + res[i]+ " "+"<br/>");
            $("#response").append(res[i]+ " "+"<br/>");
        }
    }
}
});

function Addvoterface()
{
    var vname=document.getElementById("vname").value;
    var email=document.getElementById("email").value;
    var form = new FormData();
    //var blob = dataURItoBlob(datauri)
    file = dataURLtoFile(datauri, "verify.png");
    form.append("name", vname);
    form.append("email", email);
    form.append("file", file);

```

```

var settings = {
  "url": "http://localhost:5001/register",
  "method": "POST",
  "timeout": 0,
  "processData": false,
  "mimeType": "multipart/form-data",
  "contentType": false,
  "data": form
};
$.ajax(settings).done(function(response){console.log(" ---py-res--- "+response);

});
}
function dataURLtoFile(dataurl, filename) {
var arr = dataurl.split(','), mime = arr[0].match(/:(.*?);/)[1], bstr = atob(arr[1]),
n = bstr.length, u8arr = new Uint8Array(n);
while (n--) {
u8arr[n] = bstr.charCodeAt(n);
}
return new File([ u8arr ], filename, {
type : mime
});
}
</script>
<script>
  Webcam.set({
    width: 250,
    height: 250,
    image_format: 'jpeg',
    jpeg_quality: 90
  });
  Webcam.attach( '#my_camera' );
  function take_snapshot() {
    Webcam.snap( function(data_uri) {

```

```

        $(".image-tag").val(data_uri);
        datauri=data_uri;
document.getElementById('blah').src=data_uri;
    } );
    }
//load file input image as base64 to img for preview
function previewFile() {
    const preview = document.querySelector('img');
    const file = document.querySelector('input[type=file]').files[0];
    const reader = new FileReader();
    reader.addEventListener("load", () => {
        // convert image file to base64 string
        preview.src = reader.result;
        $(".image-tag").val(reader.result);
        datauri=reader.result;
    }, false);
    if (file) {
        reader.readAsDataURL(file);
    }
}
</script>
<a href="#" id="toTop" style="display: block;"><span id="toTopHover"
style="opacity: 1;"></span></a>
        <!--end scroll_top_btn -->
<!-- for bootstrap working -->
        <script type="text/javascript" src="assets/js/bootstrap-
3.1.1.min.js"></script>
<!-- //for bootstrap working -->
</body>
</html>
//voting
<!DOCTYPE HTML>
<?php
require 'util.php';

```

```

//session_start();
if(!isset($_SESSION['username'])){
    header('Location: './user.php');
}
$test = $_SESSION['username'];
$sql = "select * from voterinfo where vname='".$_$test."";
//echo $sql;
$res = $db->query($sql);
while($row = $res->fetch_assoc()) {
    $vname = $row["vname"];
    $status = $row["status"];
    $baddress = $row["baddress"];
    $email = $row["email"];
}
?>
<html>
<head>
<title>Election a Society and People Category </title>
<!-- for-mobile-apps -->
<meta name="viewport" content="width=device-width, initial-scale=1">
<meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
<meta name="keywords" content="Election Responsive web template,
Bootstrap Web Templates, Flat Web Templates, Android Compatible web
template,
Smartphone Compatible web template, free webdesigns for Nokia, Samsung,
LG, SonyEricsson, Motorola web design" />
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/2.1.1/jquery.min.js"></script>
<script type="application/x-javascript"> addEventListener("load", function() {
setTimeout(hideURLbar, 0); }, false);
function hideURLbar(){ window.scrollTo(0,1); } </script>
<!-- //for-mobile-apps -->
<link href="assets/css/bootstrap.css" rel='stylesheet' type='text/css' />
<!--<link

```



```

href='//fonts.googleapis.com/css?family=Raleway:400,100,200,300,500,600,700,80
0,900' rel='stylesheet' type='text/css'>-->
<!--<link
href='//fonts.googleapis.com/css?family=Open+Sans:400,300,300italic,400italic,6
00,600italic,700,700italic,800,800italic' rel='stylesheet' type='text/css'>-->
<link rel="stylesheet" type="text/css" href="assets/css/style.css">
<!--strat-slider---->
<script src="mfs100-9.0.2.6.js"></script>
<!--scripttype="text/javascript" src="assets/js/adapter.min.js"></script>
<script type="text/javascript" src="assets/js/vue.min.js"></script>
<script type="text/javascript" src="assets/js/instascan.min.js"></script-->
<script
src="https://cdnjs.cloudflare.com/ajax/libs/web3/1.8.1/web3.min.js"></script>
<scriptsrc="https://cdn.jsdelivr.net/npm/truffle-contract@4.0.16/dist/truffle-
contract.js"></script>
<!--script src="js/app.js"></script-->
<script src="https://code.jquery.com/jquery-2.2.4.min.js" integrity="sha256-
BbhdLvQf/xTY9gja0Dq3HiwQF8LaCRTXxZKRutelT44="
crossorigin="anonymous"></script>
<scripttype="text/javascript"src="assets/js/jquery.blockUI.js"></script>
<script type="text/javascript" src="config.js"></script>
<!--//-slider---->
<style>
    .check
{
    opacity:0.5;
    color:#996;
}
</style>
<script >
    console.log(gOptions.contractaddress);
    function processResult(str){
        //alert(str);
        $xmlDoc = $.parseXML( str );

```

```

        //console.log(gOptions.contractaddress);
        uid = $xmlDoc.children[0].getAttribute('uid');
        $("#aid").val(uid);
    }
    var ename;
    var etype;
    var esub;
    var aadharno;
    var file;
    var candidateid="";
    var flag =0;
    var quality = 60; //(1 to 100) (recommanded minimum 55)
    var timeout = 10; // seconds (minimum=10(recommanded), maximum=60,
unlimited=0 )
</script>
<script>
    $( function() {
        //aadhar serach
        $("#aadSearch").click(function(){
            aid = $("#aid").val();
            if(!aid){
                alert('Please enter your Aadhar Number');
                return;
            }
            //get voter details using aadhar
            $.getJSON("getVDetail.php?aid="+aid, function(result){
err = result['error'];
                if(err){
                    alert(err);
                    return;
                }
                elec=result['election'];
console.log("election:"+elec);
                for(i=0;i<elec.length;i++){

```

```

        elrow=elec[i];
        console.log(elec[i]);
        $("#eldiv").css('display','block');
        $("#elname").html(elrow['ename']);
        ename=elrow['ename'];
        $("#eldate").html(elrow['edate']);
        $("#elcons").html(elrow['subtype']);
        //-----
        etype=elrow['etype']; //used in cast vote
        esub = elrow['subtype']; //used in cast vote
        $("#area").val(elrow['subtype']);
    }
    console.log(i);
    if(ename!=0){
        $("#confirmdiv").css("display","block");
        document.getElementById("btnCapture").disabled=false;
        //getcandidates(ename);
    }
});
});
</script>

</head>
<body>
<!-- header -->
<div class="header_bg">
<div class="container">
<!--start-header-->
<div class="header">
<nav class="navbar navbar-default">
<!-- Brand and toggle get grouped for better mobile display -->
<div class="navbar-header">
<button type="button" class="navbar-toggle collapsed" data-toggle="collapse"

```

```

data-target="#bs-example-navbar-collapse-1">
<span class="sr-only">Toggle navigation</span>
<span class="icon-bar"></span>
<span class="icon-bar"></span>
<span class="icon-bar"></span>
</button>
</div>
<!-- Collect the nav links, forms, and other content for toggling -->
<div class="collapse navbar-collapse nav-wil" id="bs-example-navbar-collapse-
1">
<ul class="nav navbar-nav">
<li><a href="index.php">Home</a></li>
  <li class="act"><a href="voting.php">Voting</a></li>
  <li><a href="results.php">Results</a></li>
  <li><a href="logout.php">Logout</a></li>
</ul>
</div><!-- /.navbar-collapse -->
</nav>
</div>
</div>
</div>
<div class="header_bottom">
</div>
<div class="typo">
<div class="container">
<h3 class="title">Welcome for Online Voting</h3>
<p class="nihil">Vote For Real Government.</p>
<div class="row">
<div class="col-md-9">
<h3 class="hdg">Please enter your Aadhar Number</h3>
<h4id="h4.-bootstrap-heading"><b>Adhar          Number</b></h4><input
type="text" style="border: none; border-bottom: 2px solid steelblue;width:
60%;height:50px" id="aid"/><span style="font-size: 12px" class="glyphicon
glyphicon-search" id="aadSearch"></span>

```

```

<br></br>
<br></br>
<h4 id="h4.-bootstrap-heading"><b>Constituency</b></h4>
<input type="text" readonly style="border: none; border-bottom: 2px solid
steelblue;width: 60%;height:50px" id="area"/></td>
<br></br>
<br></br>
<h4 id="h4.-bootstrap-heading"><b>Election</b></h4>
<div style="float:left" id="elname"></div>
<br></br>
<h4 id="h4.-bootstrap-heading"><b>Date :</b></h4>
<div><span id="eldate"></span></div>
<!--div style="width:40%">
    Constituency :<span id="elcons"></span>
</div-->
</div>
<div class="col-md-3">
<h3>Cameras</h3>
<div id="container">
<video width="250" height="250" autoplay="true" id="video">
</video>
</div>
<!--button onclick="capture()">Capture</button-->
<input type="button" id="btnCapture" value="Capture" disabled="true"
class=" capturebuttonpadding btn btn-primary btn-lg submit_bottom_padding"
style="margin-top:20px"/>
<br> </br>
<canvas id="canvas" style="overflow:auto"></canvas>
<input id="btnConfirm" value="Confirm" type="hidden" class="
capturebuttonpadding btn btn-primary btn-lg submit_bottom_padding"
style="margin-top:20px"/>
<br></br>
<a id="link"></a>
</div>

```

```

</div>

<form action="" method="post">
  <br>
<div class="row" style="height: 170px;display: none" id="chpersondiv">
<h3>Choose a Person</h3>
<!--Dynamically generate candidate list using js-->
</div>
<br>
<pclass="one"><divstyle="backgroundcolor:steelblue;color:white      ;height:
50px;width: 100%;border: 2px solid steelblue;"></div>
  <!--input type="submit" value="CLICK HERE TO POLL YOUR VOTE"
style="background-color:steelblue;color:white      ;height:      50px;width:
100%;border: 2px solid steelblue;"--></p>
</form>
</div>
<!-- /.row -->
<div id="wrapperProposals">
  </div>
</div>
<!-- //Vote Casting -->
<script type="text/javascript">
  $(document).ready(function() {
    $('#yes').click(function() {
      console.log("Aadhar "+aadharno);
      console.log("ename " +ename);
      console.log("candidate "+candidateid);
      // update the block message
      $.blockUI({ message: "<h1>Registering your Vote</h1>" }
        $.getJSON('./contract/Votingsol.json', function (data) {
console.log("contract");
console.log(data.abi);
const web3 = new Web3();
web3.setProvider(new
Web3.providers.WebsocketProvider('ws://localhost:8545'));

```

```

        web3.eth.net.isListening()
        .then(() => {
            console.log('is connected');
PNBContract=newweb3.eth.Contract(data.abi,gOptions.contractaddress);

var baddress = '<?php echo $baddress; ?>'
PNBContract.methods.vote(candidateid,ename).send(
{from:baddress},function(err, res){
            console.log("inside voting");
        if (err) {
            console.log("error in set!")
        }
            else
            {
                console.log(res);
                AddVote(res);
            }
        });
    })
    .catch(e => {
        console.log('Wow. Something went wrong: ' + e);
        alert("Ganache Server not Running");
    });
});

function AddVote(res) {
    var txhash=res;
    url1="castVote.php";
data1={'aadhar':aadharno,'ename':ename,'candidate':candidateid,'etype':etype,
'esubtype':esub,'txhash':txhash};
    $.post(url1,data1,function(){},'json')
        .done(function (response){
            console.log(response.status);
//sendMail(txhash); //enable to send mailalert ("Successfully Voted");

```

```

        location.href = './index.php';
    }).fail(function (jqxhr, textStatus, error){
        var err = textStatus + ", " + error;
        console.log("Request Failed: " + err);
    });
}

$('#no').click(function() {
    $.unblockUI();
    return false;
});

//function to capture the face.

$("#btnCapture").click(function() {
    var canvas = document.getElementById('canvas');
    var video = document.getElementById('video');
    canvas.width = 250;
    canvas.height = 250;
    canvas.getContext('2d').drawImage(video, 0, 0, 250,250);
    document.getElementById("btnConfirm").type="button";
    file = dataURLtoFile(canvas.toDataURL('image/jpeg'), "verify.png");
    //var res = getCandidates();
});

function dataURLtoFile(dataurl, filename) {
var arr = dataurl.split(','); mime = arr[0].match(/:(.*?);/)[1], bstr = atob(arr[1]),
n = bstr.length, u8arr = new Uint8Array(n);
while (n--) {u8arr[n] = bstr.charCodeAt(n);
}
return new File([ u8arr ], filename, {
type : mime
});
}

//function to confirm
$("#btnConfirm").click(function(){
    aid = $("#aid").val();
    aadharno=aid;

```



```

if (ename != "")
{
var form = new FormData();
    form.append("name", file.name);
    form.append("file", file);
    var settings = {
        "url" : "http://localhost:5001/",
        "method" : "POST",
        "timeout" : 0,
        "processData" : false,
        "mimeType" : "multipart/form-data",
        "contentType" : false,
        "data" : form
    };
    $.ajax(settings).done(function(response) {
        console.log("---py-res---"+response);
        console.log(response.status);
        var obj=JSON.parse(response);
        console.log(isEmpty(obj));
        if(!isEmpty(obj))
        {
            if (obj[0].status)
            {
                getCandidates();
            }
            else
            {
                alert("Voter face not matched");
            }
        }
        else
        {
            alert("No Proper Face Detected");
        }
    }

```

```

        }).fail(function (jqXHR, textStatus) {
            console.log(jqXHR);
        });
    }
});

function isEmpty(obj) {
    return Object.keys(obj).length === 0;
}

function getCandidates()
{
    $.getJSON('./contract/Votingsol.json', function (data) {
        console.log("contract");
        console.log(data.abi);
    });

    var web3 = new Web3(new Web3.providers.HttpProvider('http://localhost:8545'));
    web3.eth.net.isListening()
        .then(() => {
            console.log('is connected');
        });

    PNBContract = new web3.eth.Contract(data.abi, contractAddress);
    PNBContract.methods.getCandidates(ename).call(function (err, res) {
        if (res) {
            console.log("insidename");
            console.log(res.length);
            var out = res;
            console.log(res);
            var iterator = res.values();

            $("#chpersondiv").css("display", "block");
            $("#chpersondiv").css("position", "relative");

            //console.log(iterator);
            for (let elements of iterator) {
                if (elements !== "")
                {
                    var text = elements;

                    const myArray = text.split("-");

```

```

var candidate = myArray[0];
var symbol = myArray[1];
var candidateid=myArray[2];
    document.getElementById('chpersondiv').style.height='150px';
    var inputTag = document.createElement("input");
    inputTag.setAttribute("id", candidateid);
    inputTag.setAttribute("type", "button");
inputTag.setAttribute("style", "height:100px;width:150px;           background-
size:cover;                                           background-
image:url('../An_OnlineVotingQR_Admin/upImages/'+symbol+'
inputTag.onclick=function(){your_function_name(this.id)};
                var div = document.createElement("div");
                inputTag.setAttribute("value", candidate);
                inputTag.innerHTML=candidate;
document.getElementById('chpersondiv').appendChild(inputTag);
                var label = document.createElement("label");
                label.style.marginRight = "15px";
document.getElementById('chpersondiv').appendChild(label);
                }
            }
        }
        if(err)
        {
            console.log(err);
        }
    });
    })
    .catch(e => {
        console.log('Wow. Something went wrong: '+ e);
        alert("Ganache Server not Running");
    });
    });
}
functionyour_function_name(clicked){$(".imgcheck").removeClass("check");$.

```

```

blockUI({ message: $('#question'), css: { width: '275px' } });

candidateid=clicked;

$(this).parent().children("img").toggleClass("check");
        console.log('candidate id'+candidateid);
    }

//send mail after successfull voting
function sendMail(res)    {
    var data = {
        'vname': '<?php echo $vname; ?>',
        'email': '<?php echo $email; ?>',
        'hash' : res
    };

    $.ajax({
        url: 'mailto.php',
        data: data,
        type: 'POST',
        success: function (data) {
            res = $.parseJSON(data);
            console.log("message"+res);
            console.log(res['message']);
            /*if(res['status']){
                console.log(data.error);
            }
            else
            {
                console.log(data.message);
            }*/
        }
    })
}

</script>
<script>
var video = document.querySelector("#video");

```

```

if (navigator.mediaDevices.getUserMedia) {
    navigator.mediaDevices.getUserMedia({ video: true })
        .then(function (stream) {
            video.srcObject = stream;
        })
        .catch(function (error) {
            console.log("Something went wrong!");
        });
}
</script>
<div id="question" style="display:none; cursor: default">
    <h1>Confirm to Cast your Vote.</h1>
    <input type="button" id="yes" value="Confirm" />
    <input type="button" id="no" value="No" />
</div>
<!-- footer -->
    <div class="footer">
        <div class="container">
<div class="footer-copy">
<p>All rights reserved | www.onlineelectionvoting.com</a></p>
        </div>
    </div>
</div>
<!-- //footer -->
    <!-- scroll_top_btn -->
<script type="text/javascript" src="assets/js/move-top.js"></script>
<script type="text/javascript" src="assets/js/easing.js"></script>
    <script type="text/javascript">
        $(document).ready(function() {
            var defaults = {
                containerID: 'toTop', // fading element
idcontainerHoverID: 'toTopHover', // fading element hover id
                scrollSpeed: 1200,
                easingType: 'linear'
            }
        });
    </script>

```

```

        };
        $().UiToTop({ easingType: 'easeOutQuart' });
    });
</script>

<a href="#" id="toTop" style="display: block;"><span id="toTopHover"
style="opacity: 1;"></span></a>

<!--end scroll_top_btn -->

<!-- for bootstrap working -->
<!--scripttype="text/javascript"src="assets/js/camera/app.js"></script-->
<script type="text/javascript" src="assets/js/bootstrap-3.1.1.min.js"></script>

<!-- //for bootstrap working -->
</body>
</html>
//logout
<?php
session_start();
session_destroy();
header('Location: user.php');
exit;
?>
//Dshboard
<!DOCTYPE HTML>
<?php
require 'util.php';
//session_start();
if(!isset($_SESSION['username'])){
    header('Location: './user.php');
}
$test = $_SESSION['username'];
$sql = "select * from voterinfo where vname='".$test."'";
//echo $sql;
$res = $db->query($sql);
while($row = $res->fetch_assoc()) {

```

```

        $status = $row["status"];
    }
?>
<html>
<head>
<style>
.button {
    background-color: #4CAF50;
    border: none;
    color: white;
    padding: 15px 32px;
    text-align: center;
    text-decoration: none;
    display: inline-block;
    font-size: 16px;
    margin: 4px 2px;
    cursor: pointer;
}
</style>
<title>Election a Society and People Category </title>
<!-- for-mobile-apps -->
<meta name="viewport" content="width=device-width, initial-scale=1">
<meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
<meta name="keywords" content="Election Responsive web template,
Bootstrap Web Templates, Flat Web Templates, Android Compatible web
template,
Smartphone Compatible web template, free webdesigns for Nokia, Samsung,
LG, SonyEricsson, Motorola web design" />
<script type="application/x-javascript"> addEventListener("load", function() {
setTimeout(hideURLbar, 0); }, false);
function hideURLbar(){ window.scrollTo(0,1); } </script>
<!-- //for-mobile-apps -->
<link href="assets/css/bootstrap.css" rel='stylesheet' type='text/css' />
<!--<link

```

```
href="//fonts.googleapis.com/css?family=Raleway:400,100,200,300,500,600,700,800,900" rel='stylesheet' type='text/css'>-->
<!--<link
href="//fonts.googleapis.com/css?family=Open+Sans:400,300,300italic,400italic,600,600italic,700,700italic,800,800italic" rel='stylesheet' type='text/css'>-->
<link rel="stylesheet" type="text/css" href="assets/css/style.css">
<!--strat-slider---->
<scripttype="text/javascript"src="assets/js/jquery-1.11.1.min.js"></script>
<!--//-slider---->
</head>
<body>
<!-- header -->
    <div class="header_bg">
        <div class="container">
            <!--start-header---->
            <div class="header">
                <div class="logo">
<!-- <a href="index.php"></a> -->
                </div>
<nav class="navbar navbar-default">
<!-- Brand and toggle get grouped for better mobile display -->
<div class="navbar-header">
<button type="button" class="navbar-toggle collapsed" data-toggle="collapse"
data-target="#bs-example-navbar-collapse-1">
<span class="sr-only">Toggle navigation</span>
<span class="icon-bar"></span>
<span class="icon-bar"></span>
<span class="icon-bar"></span>
</button>
</div>
<!-- Collect the nav links, forms, and other content for toggling -->
<div class="collapse navbar-collapse nav-wil" id="bs-example-navbar-collapse-1">
<ul class="nav navbar-nav">
```

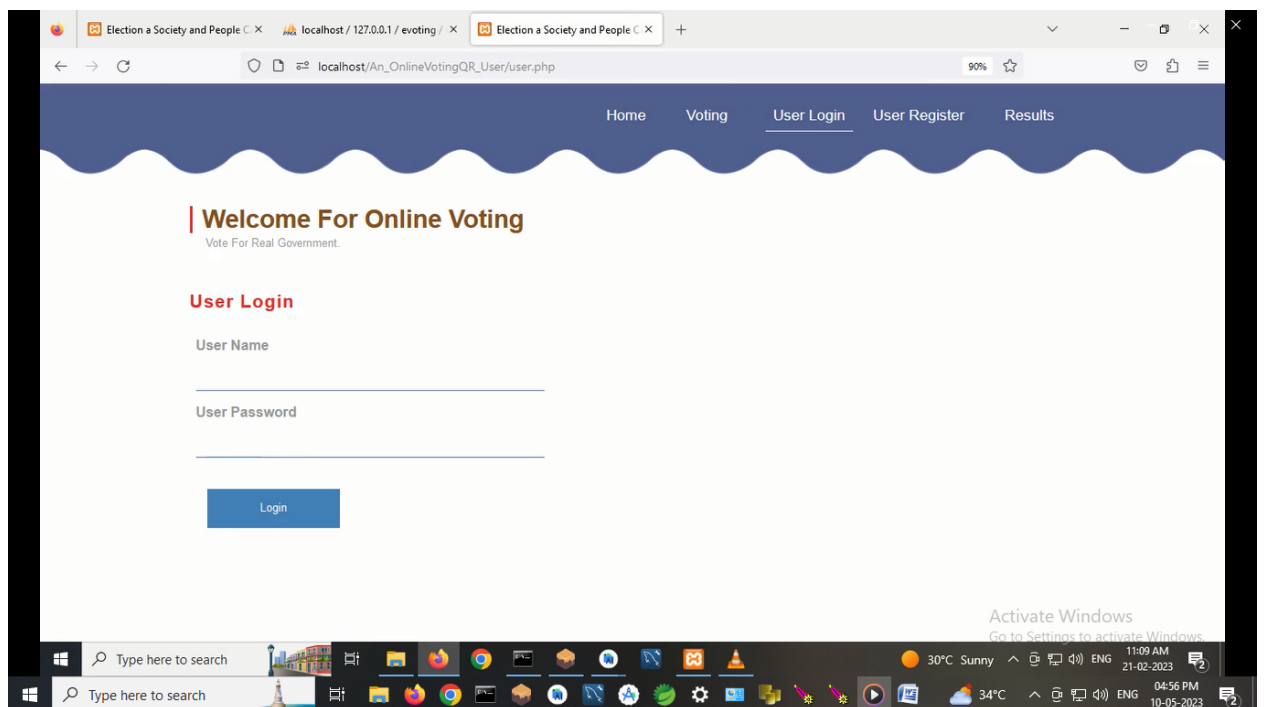
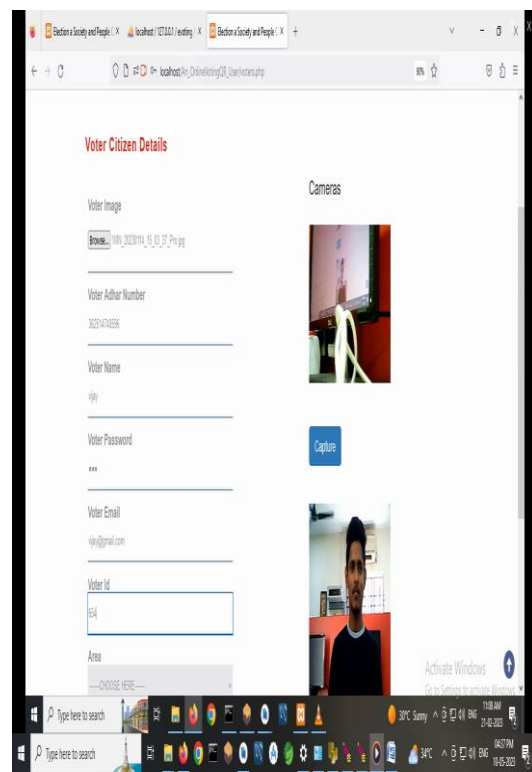
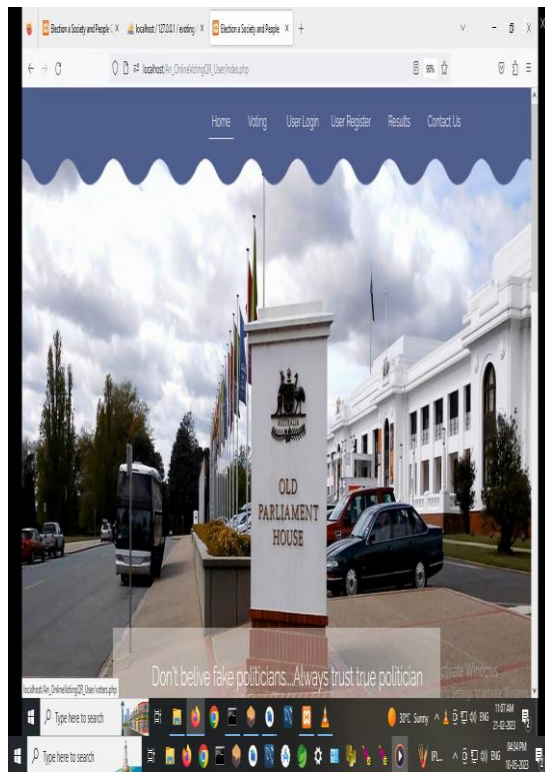


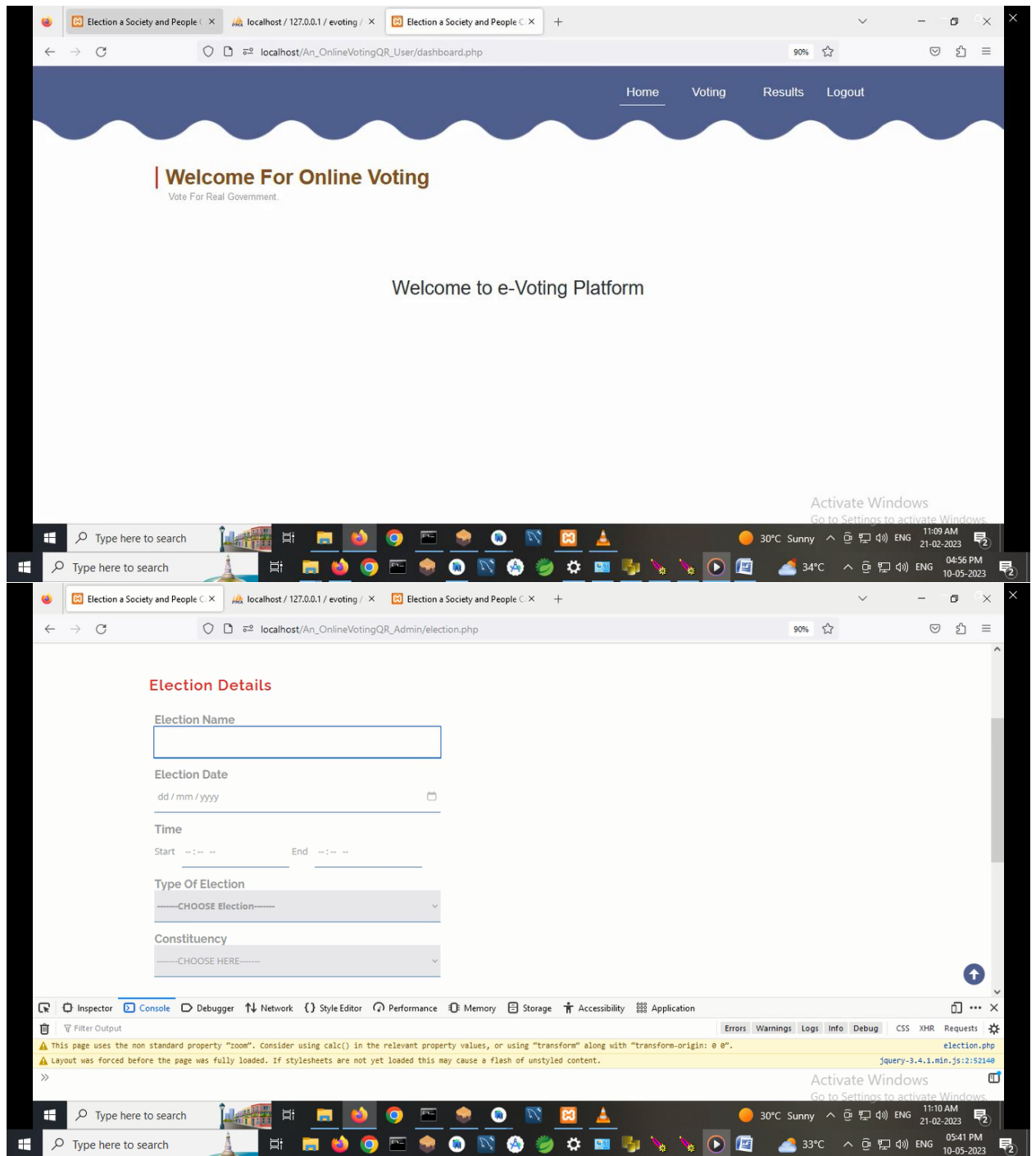
```
<li class="act"><a href="index.php">Home</a></li>
  <!--<li><a href="admin.php">Login</a></li>-->
  <li><a href="voting.php">Voting</a></li>
  <li><a href="results.php">Results</a></li>
  <li><a href="logout.php">Logout</a></li>
</ul>
</div><!-- /.navbar-collapse -->
</nav>
</div>
</div>
</div>
<div class="header_bottom">
</div>
<!-- //end-header -->
<!-- banner1 -->
<!-- //banner1 -->
<!--typography-page -->
<div class="typo">
<div class="container">
<h3 class="title">Welcome for Online Voting</h3>
<p class="nihil">Vote For Real Government.</p>
<?php
if($status == "Pending"){
?>
<br/>
<br/>
<br/>
<br/>
<br/>
<h2><center>Your Request Is Still Under Verification Process</center><h2>
<?php
}
else if($status == "Accepted"){
?>
```

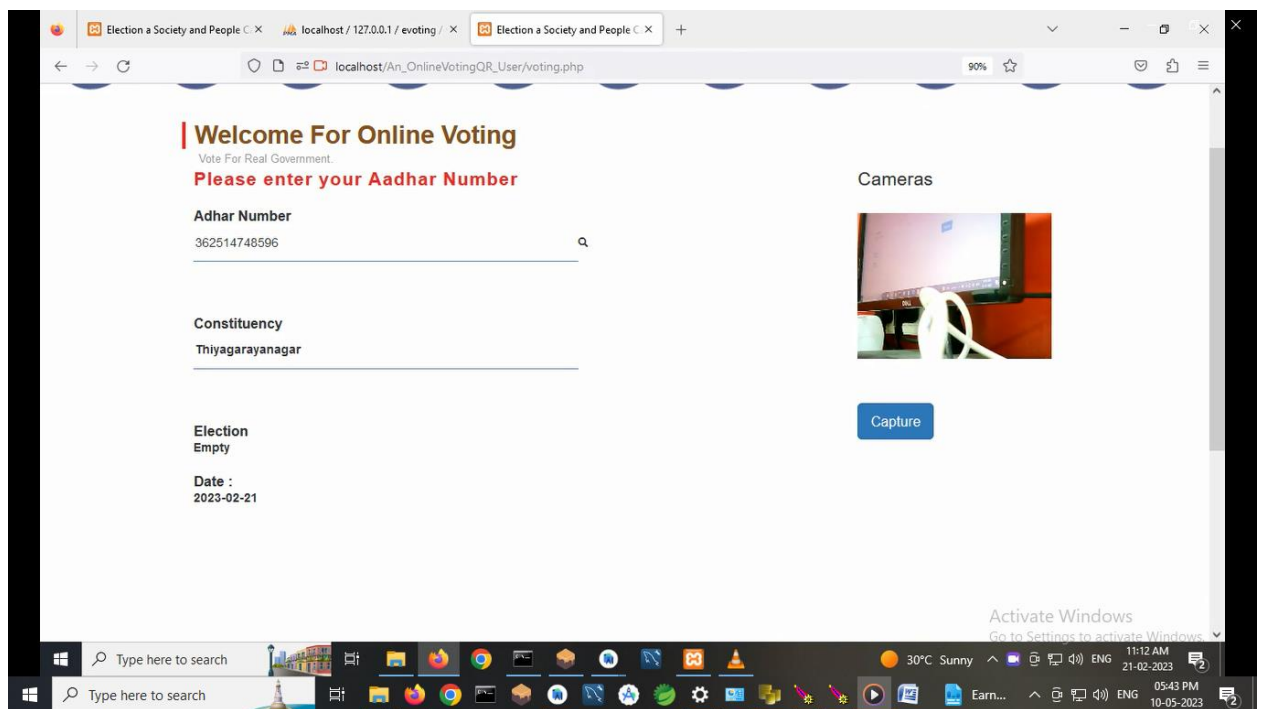
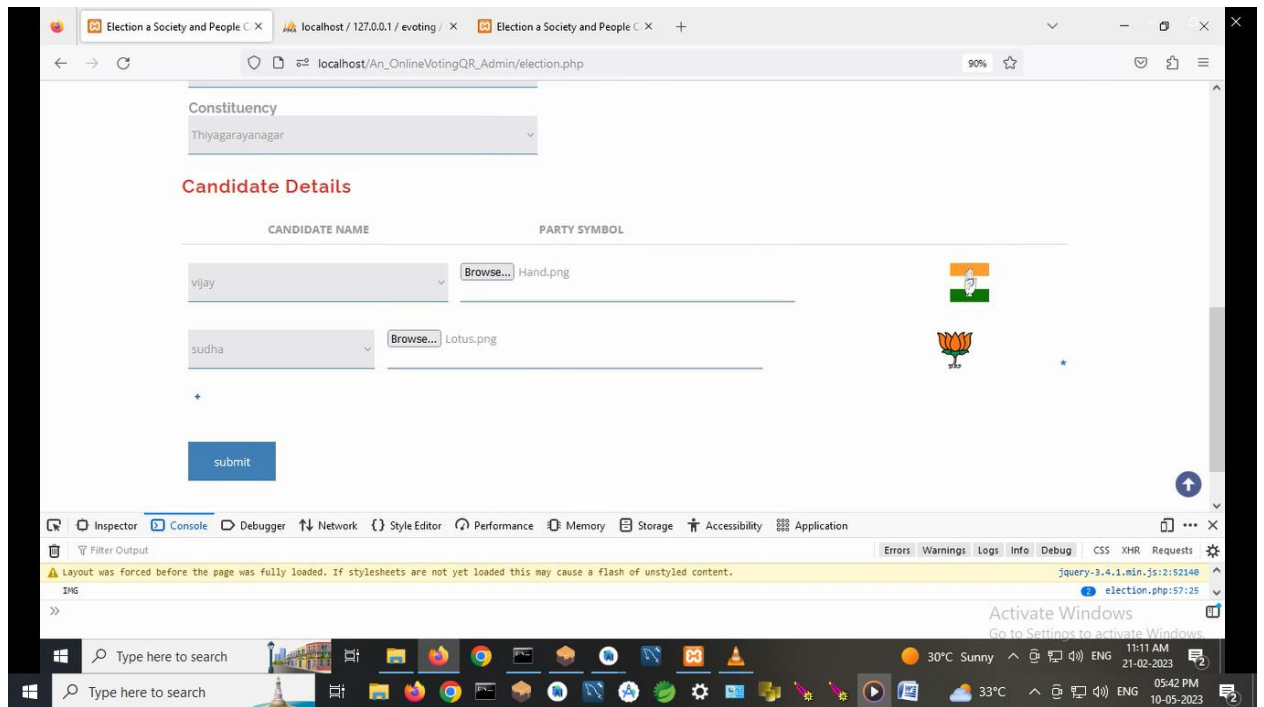
```
<br/>
<br/>
<br/>
<br/>
<br/>
<h2><center>Welcome to e-Voting Platform</center></h2>
<?php
}
else
{
?>
<br/>
        <br/>
        <br/>
        <br/>
        <br/>
        <h2><center>Your Request Has Been Rejected Please
Submit Your detail Again</center></h2>
        <br/>
        <br/>
        <br/>
        <center><div style="background-color: steelblue;color:
white;width: 20%;font-size: 30px;text-align: center;height: 50px;">
        <a href="votersupdate.php" style="color: white">Update detail</a>
        </div></center>
<?php
}
?>
</div>
</div>
</div>
<!-- //typography-page -->
<!-- scroll_top_btn -->
<script type="text/javascript" src="assets/js/move-top.js"></script>
```

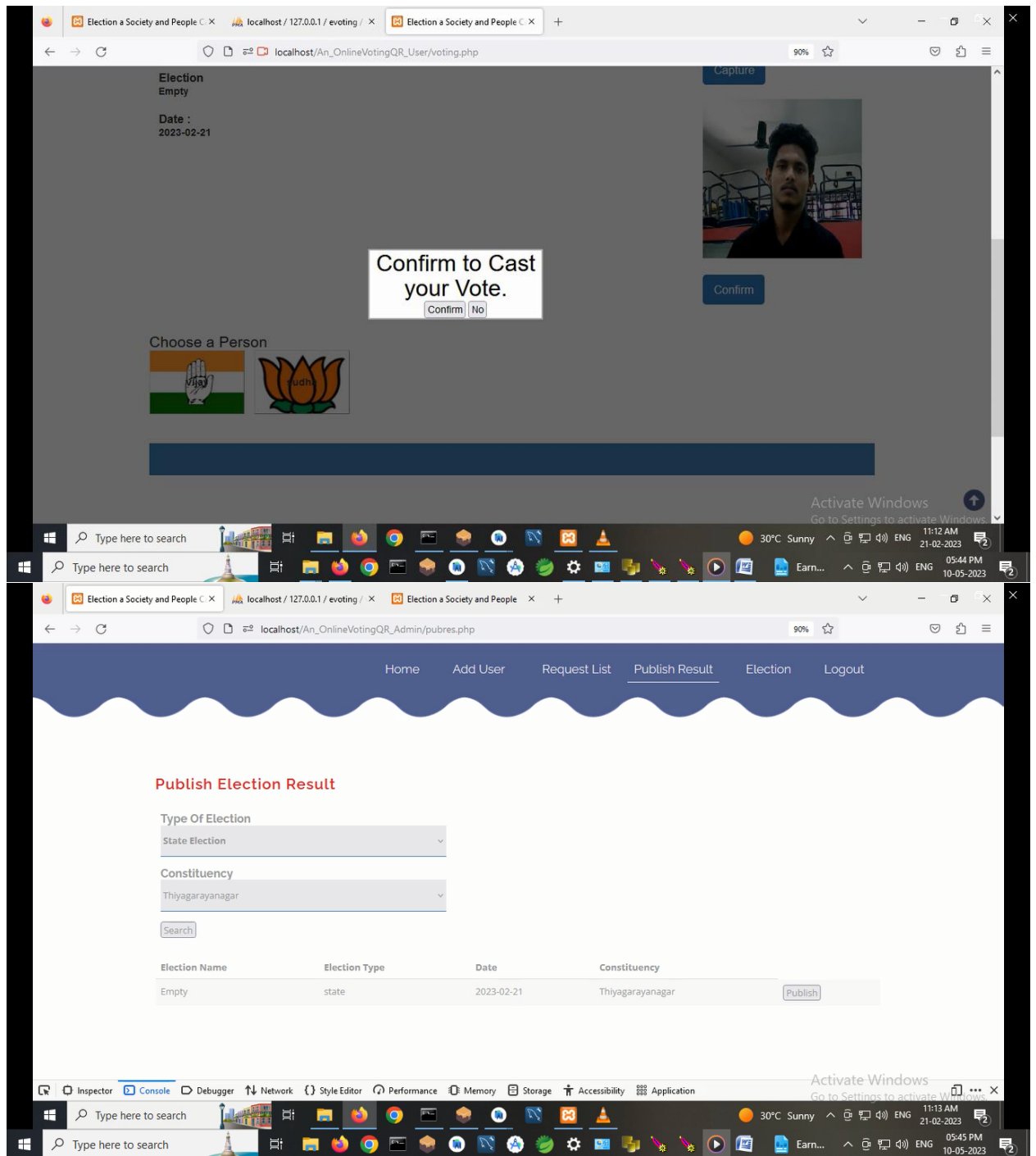
```
<script type="text/javascript" src="assets/js/easing.js"></script>
<script type="text/javascript">
$(document).ready(function() {
var defaults = {
containerID: 'toTop', // fading element id
containerHoverID: 'toTopHover', // fading element hover id
scrollSpeed: 1200,
easingType: 'linear'
};
$.UItoTop({ easingType: 'easeOutQuart' });
});
</script>
<a href="#" id="toTop" style="display: block;"><span id="toTopHover"
style="opacity: 1;"></span></a>
<!--end scroll_top_btn -->
<!-- for bootstrap working -->
<script type="text/javascript" src="assets/js/bootstrap-
3.1.1.min.js"></script>
<!-- //for bootstrap working -->
</body>
</html>
```

Screenshots:









Future Enhancement

We focus on improving the security level by guaranteeing the integrity of e-coupon information. For example, the existing e-coupon service uses a database system. In this system, since an administrator can easily obtain the authority, the administrator rather easily modifies the data maliciously. Meanwhile, in our provided service which uses a blockchain system, the administrator cannot easily obtain the authority since the authority should be obtained by the consensus of all users.

Conclusion:

We have investigated e-coupon services that store e-coupon information on a decentralized server. We found that the e-coupon information stored in the server can be manipulated by a malicious attacker or administrator. To handle this issue, we present a new e-coupon service that improves security by exploiting e-coupon smart contracts in a Ethereum blockchain system using Ganache Software.

Future Work :

Block chain-based voting, which relies on a decentralized, distributed digital ledger is vulnerable to many of the security flaws inherent in internet voting, such as the potential for malware to alter votes on a voter's local device before the ballot is transmitted and the lack of secret ballots.

Conclusion:

The purpose of proposing a block chain-based solution for the voting system was to build trust between government and voters to make-believe that their voting integrity is kept safe. The block chain-based voting is also make the voting process transparent and trustworthy. Our proposal enables a voter to cast his/her vote through internet without going to voting booth and additionally registering himself/herself for voting in advance, proxy vote or double voting is not possible, fast to access, highly secure, easy to maintain all information of voting, highly efficient and flexible. Hence, by this voting percentage will increase drastically.

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